

4.3 Contrastive Learning and Zero-Shot Transfer

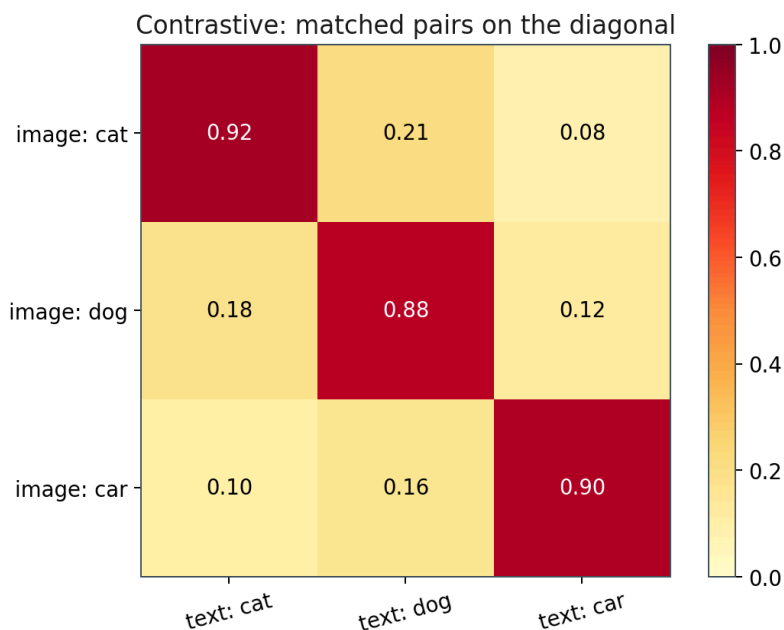
InfoNCE wants the diagonal. Zero-shot is nearest prompt, not a trained C-way head.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Contrastive learning pulls matched pairs together and pushes unmatched pairs apart, in a shared space. **Zero-shot** transfer is what you get when one side of the pair can be a class name written as text. This is a **coordination** loss: two towers, a similarity, no fused decoder. Week 5 will put CLIP's data and retrieval on the table. This note is the loss and the zero-shot trick.

1. What contrastive learning and zero-shot classification are

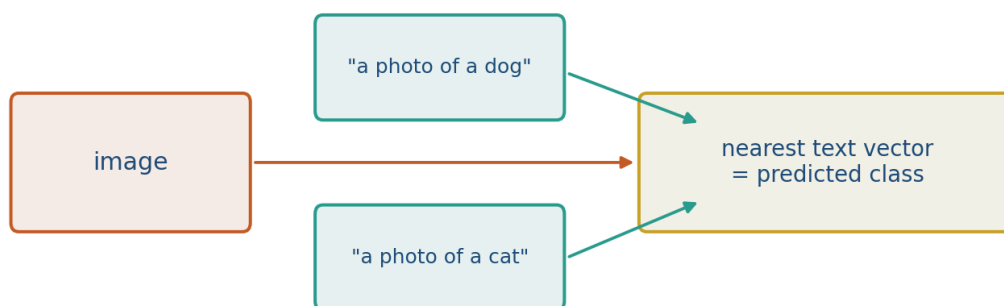
Contrastive learning pulls matched pairs together and pushes the rest apart. Take N images and N captions. Encode both. Cosine similarities form an $N \times N$ matrix. The loss (InfoNCE; Oord et al. 2018) wants the **diagonal** hot: image i with caption i , not caption j .



This is coordinated representation (note **4.1**) with a specific loss. You never fuse pixels and words into one vector during pretraining; you only score pairs. SimCLR is the vision-only cousin (two crops of the same image). We cite it; we do not assign it.

Zero-shot classification is: classify by nearest prompt, not a C -way trained head. At test time, encode the image once. Encode prompts "a photo of a dog", "a photo of a cat", and so on. Take the nearest prompt. There is no trained softmax over ImageNet. New classes are new strings.

Zero-shot: class names are text embeddings, not trained heads



CLIP was trained on **phrases**. The raw label "dog" often loses to "a photo of a dog". Prompt **ensembling** (average several phrasings) is a test-time trick, not a new model. We mention it; Lab 4 does not require it.

A class the text tower never saw as a phrase will not magically appear. Open-vocabulary is not infinite-vocabulary.

2. Why we use it

You may not have labels for every class at train time. A similarity space transfers to new names: adding a class is adding a string, not training a new column of a softmax. That is why CLIP-style models are called foundation models for vision–language, and why this course treats contrastive learning as a method rather than a trick.

A C -way head is the older default. It works when you have a labeled set for those C classes. It fails when the next user wants a class you did not train. Zero-shot is the transfer story that does not need that labeled set.

Web pairs are noisy (Week 5: BLIP filters them). Contrastive spaces can still be **biased** (Week 5.3). The loss does not make the space fair; it only makes matched pairs closer than unmatched ones.

3. Architecture

Two encoders (or two views of one network). A batch matrix of scores. Softmax over the row (or both directions).

Image i maps to $v_i = f(x_i)$. Text j maps to $t_j = g(y_j)$. The interaction is cosine, not a concatenating MLP. CLIP uses **both** directions (image-to-text and text-to-image). Temperature τ sharpens the softmax. Small τ makes the diagonal even more peaked, but the **argmax** of a row does not change if you only rescale that row.

The negative set is the rest of the batch. Small batches $\Rightarrow$ easy negatives $\Rightarrow$ a weak space. CLIP's real batch was tens of thousands. Lab 4 is $N = 4$ so you can see the matrix, not so you can match

ImageNet.

At test time the architecture does not grow a classification head. The classifier **is** the text tower, queried with class phrases.

4. How it works, step by step

Lab 4: four pairs, heatmap of the $N \times N$ scores. The diagonal should win.

1. **Encode a batch.** N matched (x_i, y_i) pairs. Image tower and text tower.
2. **Fill the matrix.** $s_{ij} = \cos(v_i, t_j)$. The diagonal is the true matches; the off-diagonal entries are **negatives**, not "don't care."
3. **InfoNCE on each row.** Softmax over captions for a fixed image; the loss is $-\log$ of the diagonal probability. CLIP also does the other direction.
4. **Zero-shot at test.** Freeze both towers. Embed the image. Embed one prompt per class. Predict $\arg \max_c \cos(f(x), g(\text{prompt}_c))$.
5. **Do not add a class the text tower cannot spell.** Open-vocabulary is not infinite-vocabulary.

If every caption is the same vector, every row of S is constant, softmax is uniform, and the diagonal is not special: **the loss cannot learn matching**.

5. Mathematical formulas

Let $s_{ij} = \cos(u_i, v_j)$. InfoNCE, one direction, averaged over the batch:

$$\mathcal{L} = -\frac{1}{N} \sum_i \log \frac{\exp(s_{ii} / \tau)}{\sum_j \exp(s_{ij} / \tau)}.$$

Row i alone is

$$-\log \frac{\exp(\text{sim}(v_i, t_i) / \tau)}{\sum_{j=1}^N \exp(\text{sim}(v_i, t_j) / \tau)}.$$

Zero-shot: embed class phrases, pick

$$\hat{c} = \arg \max_c \cos(f(x), g(\text{prompt}_c)).$$

Dividing the cosines by τ before a softmax changes the distribution, not the argmax of a single row of scores.

6. Positive points and negative points

Positive.

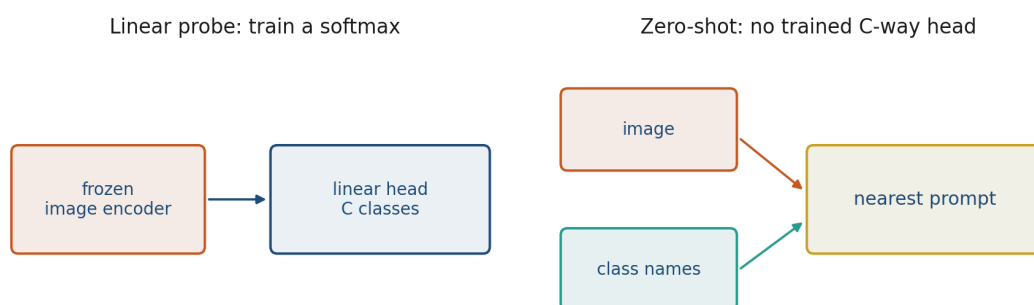
- No C -way head at train time; new classes can be new prompts.
- The same space supports retrieval (rank by cosine) and zero-shot classification.
- Lab 4's $N = 4$ grid is the whole method in miniature.

Negative.

- Needs big batches so the negatives are hard enough.
- Temperature τ is a knob: it sharpens, it does not change argmax on one row.
- Compositionality is not free; a batch of 4 will not teach "mug in grass" versus "grass in a mug."
- A class the text tower never saw as a phrase will not appear at test time.

When not to. If you already have a labeled set for a fixed C and you only need those C classes, a linear probe on a frozen encoder can be the tighter transfer. Zero-shot is for names you did not train as a head.

7. Linear probe versus zero-shot



Linear probe: freeze the image encoder, train a C -way softmax on labeled images. Needs a training set for those C classes.

Zero-shot: no extra weights. The classifier is the text tower. Adding a class is adding a string.

Both are transfer. Zero-shot is what makes CLIP a **foundation** model. Captioning, LLaVA, and Flamingo wait for later weeks.

8. Teaching this note

~18 minutes. Draw the $N \times N$ grid, mark the diagonal, write InfoNCE for $N = 2$, then swap the text side for class prompts, then probe vs zero-shot. Play the CLIP video on **zero-shot** and **the similarity matrix** (9:00–22:25). Lab 4's tiny InfoNCE is this grid with $N = 4$.

9. Worked example

Two pairs. Cosine matrix (already normalized):

$$S = \begin{bmatrix} 0.9 & 0.1 \\ 0.2 & 0.8 \end{bmatrix}, \quad \tau = 1.$$

Row 0 softmax: $e^{0.9} \approx 2.46$, $e^{0.1} \approx 1.11$, so $[0.69, 0.31]$. InfoNCE term $-\log 0.69 \approx 0.37$.

InfoNCE: the rest of the batch are negatives. The diagonal must be special.

row 0 scores	softmax	$-\log p_{\text{diag}}$
$[0.9, 0.1], \tau = 1$	$[0.69, 0.31]$	0.37
identical captions	uniform	cannot learn

If every caption is the same vector, every row of S is constant, softmax is uniform, and the diagonal is not special: **the loss cannot learn matching**.

Zero-shot: image $v = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$, prompts $\text{dog} = \begin{bmatrix} 0.9 \\ 0.1 \end{bmatrix}$, $\text{cat} = \begin{bmatrix} 0.1 \\ 0.9 \end{bmatrix}$. Cosines 0.99 vs 0.10. Predict dog.

Dividing by $\tau = 0.07$ before softmax does **not** change that argmax.

10. Where students get stuck

- Treating off-diagonal as “don’t care” instead of **negatives**.
 - Adding a class at test time that the text tower cannot spell.
 - Using batch size 4 in a real CLIP run and expecting ImageNet-level negatives.
 - Calling zero-shot a trained ImageNet head.
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11. Video

Watch [Yannic Kilcher: OpenAI CLIP, Connecting Text and Images](#).

Pause on **zero-shot (9:00)** — class names as text — and on **InfoNCE / the $N \times N$ matrix (14:40)**. That matrix is the whole note.

12. Practice

1. If every caption in the batch is the word *photo*, what happens to the diagonal?
2. Why is "a photo of a golden retriever" often better than "golden retriever" for CLIP-style zero-shot?
3. Name one class you **cannot** add at test time by typing it.
4. For $S = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $\tau = 1$, write the two InfoNCE row-losses (they should be 0). Now change S_{01} to 1. Recompute row 0's softmax and $-\log p_{\text{diag}}$.
5. Three prompts with cosines to one image 0.2, 0.5, 0.1. Which class wins zero-shot? If you divide the cosines by $\tau = 0.07$ before softmax, does the **argmax** change?
6. Linear probe versus zero-shot: which one needs a labeled training set for the C classes?
7. CLIP uses both image-to-text and text-to-image InfoNCE. Why is one direction not enough if you also want to retrieve images from a caption?